

# NIKHIL BAROT

AI Engineer | GenAI | Agentic AI | ML Research | Finance | Healthcare

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## SUMMARY

AI/ML engineer with **10 years** of enterprise software experience, including three years of building production GenAI and multi-agent systems.

**IEEE author** with publication on Machine Learning Research paper on a novel predictive-maintenance framework in IEEE Xplore.

Developed a **Novel Bio** inspired Neural Network which can recognize Analog patterns.

## TECHNICAL SKILLS

- **Agentic AI & Orchestration:** Lang Chain, Lang Graph, Llama-3 (70B/8B), Google ADK, Model Context Protocol, ReAct agents, Mistral, LLaVA (Multimodal), OpenAI API, RAG Pipelines, Synthetic Data Generation, Microsoft Foundry Agent
- **LLM & GenAI:** GPT-4, Gemini, Azure OpenAI, prompt engineering, RAG, NL-to-SQL, semantic search, embeddings
- **Evaluation & AI Safety:** DeepEval, RAGAS, DSPy, Guardrails AI, Microsoft Presidio, LLM regression testing
- **Backend & APIs:** Python, FastAPI, REST APIs;
- **Cloud & DevOps:** GCP (Vertex AI, Vertex AI Vector Search, Cloud Run, Big Query), Docker, Kubernetes, Terraform, Jenkins
- **Data & Vector Search:** FAISS, Pinecone, PostgreSQL
- **AI Coding Agents:** Claude Code, Cursor, GitHub Copilot (Ask & Agent modes), Gemini, ChatGPT
- **Languages:** Python, SQL, TypeScript, Java, Unix, PL SQL,

## CERTIFICATION

- Google ADK — Agent Development Kit | Udemy, 2026
- Fundamentals of Deep Learning | NVIDIA, 2025
- Generative AI Professional | Oracle, 2025
- AI Vector Search Professional | Oracle, 2025
- Data Science Professional | Oracle, 2025

## EXPERIENCE

### Central Michigan University

AI / ML Research Engineer

Mt. Pleasant, MI

August 2025 – May 2026

- **Architected Hybrid SAMB-GRU**, a regime-aware dual-head deep learning framework (Conv1D → BiGRU → Temporal Attention) for Remaining Useful Life (RUL) prediction on the NASA C-MAPSS aerospace benchmark; achieved  **$R^2 = 0.8825$  and RMSE = 13.48**, outperforming CNN-LSTM, LSTM, GRU, XGBoost, and Random Forest baselines; **published at IEEE Xplore** paper.
- **Designed a regime-aware K-Means clustering preprocessing pipeline** isolating six operational flight regimes with per-cluster normalization; removing it inflated the PHM safety penalty by up to 138%, statistically validating it as the most critical system component using ablation analysis.
- **Built unsupervised diagnostic probes** over a shared latent encoder to attribute subsystem-level fault signatures (Fan/LPC, HPC/Core, Turbine) without fault-label supervision, producing physically coherent degradation patterns across all four NASA C-MAPSS engine subsets.
- **Implemented inverse-RUL sample weighting and leakage-free validation splits** aligned with the asymmetric PHM08 safety objective; reduced FD004 penalty score to  $1,041 \pm 96.9$  vs. 2,474 baselines, demonstrating statistical rigor in uncertainty quantification and safety-critical model evaluation.
- **Invented a Neuroidal Network** for clinical ECG analysis, replaced standard perceptrons with biologically inspired neuroidal computational units, encoded arbitrary topologies via adjacency matrices, and built a fully custom non-backpropagation AI architecture with no market equivalent.
- **Trained the model entirely using Genetic Algorithms (evolutionary optimization)**, achieving **99% accuracy** on unseen ECG patient data with precise R-peak detection; designed and ran end-to-end model evaluation including confusion matrix, precision, recall, and F1 analysis for clinical-grade validation.

- **Built incremental fine-tuning and continuous model improvement pipelines:** pre-trained genetic populations reload and evolve on new patient data without full retraining, directly addressing real-world healthcare deployment requirements (model updating & performance drift management).
- **Validated cross-domain generalizability:** Tested the ECG architecture on a human audio binary classification task achieving 76% accuracy, demonstrating transferability of evaluation methodology across clinical data modalities.
- **Preprocessed and cleaned real-world ECG signal data,** handled noise, waveform normalization, and clinical feature extraction; designed data pipelines with rigorous quality controls ensuring clinical-grade model input.
- **Clinical BERT Optimization (PEFT/LoRA):** Fine-tuned Clinical BERT using LoRA adapters—freezing the pretrained encoder to achieve a 98% reduction in trainable parameters and 60% lower GPU memory utilization while boosting multi-class F1-score from 70% to 91%.

## Central Michigan University — Office of Information & Technology

*AI Engineer*

*Mt. Pleasant, MI*

*May 2025 – Aug 2025*

- **Agentic NL-to-SQL Analytics:** Architected a multi-tool agentic service using Python, FastAPI, Gemini, and Google ADK that dynamically handles schema discovery, SQL validation, and secure execution—improving query accuracy by 35% and reducing manual report generation time by 80%
- **Local SLM Fine-Tuning & Routing:** Fine-tuned a Mistral 7B model using LoRA/PEFT to build a dynamic router agent for internal support ticketing, achieving 94% triage accuracy across IT and Finance workflows.
- **Human-in-the-Loop Workflow Automation:** Engineered a Human-in-the-Loop grant underwriting agent utilizing Chain-of-Thought (Few-Shot) prompting to draft automated risk assessment memos, accelerating research grant dispersal by 30%
- **Serverless MLOps & Platform Delivery:** Containerized and deployed AI microservices on Google Cloud Run via Docker, Terraform, and GitHub Actions CI/CD pipelines, slashing deployment turnaround time by 70%.

## JPMorgan Chase & Co. — Asset & Wealth Management

*Associate — AI Engineer*

*Mumbai, India*

*Aug 2021 – Dec 2024*

- **Multi-Agent Systems & Test Automation:** Architected a stateful **LangGraph** multi-agent platform featuring conditional routing, shared memory, and **Human-in-the-Loop** validation to automate Jira analysis and BDD/Selenium suite generation, cutting manual test development effort by 60%.
- **Enterprise RAG & Knowledge Retrieval:** Engineered an end-to-end RAG pipeline using **FastAPI**, **Gemini**, and **Vertex AI Vector Search**, implementing document ingestion, intelligent chunking, semantic vector search, and metadata filtering , reducing document discovery time by **70%**.
- **LLM Evaluation Frameworks:** Implemented automated LLM evaluation suites (**DeepEval**, **RAGAS**) to benchmark reasoning, tool selection, and code generation, improving retrieval quality by 40% and reducing AI regression defects by 35% prior to release. Familiarity with benchmarks including **MMLU**, **HumanEval**, **SWE-bench**, and **BFCL**.
- **AI Safety, Governance & Observability:** Enforced enterprise AI compliance using Guardrails AI and Microsoft Presidio for prompt-injection defense and PII masking; built an LLM observability stack to track drift and token usage, reducing production incidents by 45%
- **Cloud Infrastructure & MLOps:** Automated production CI/CD pipelines for Vertex AI microservices using Docker, Terraform, and GitHub Actions, establishing repeatable deployments, automated testing, and version-controlled infrastructure.
- **Technical Leadership & Delivery:** Led and mentored an 8-engineer AI workstream, overseeing sprint planning, code reviews, and onboarding while presenting live system demos to senior stakeholders across North America, APAC and EMEA regions.

## Oracle (Client: State Street Bank)

*Staff Consultant*

*Mumbai, India*

*Oct 2018 – Aug 2021*

- Spearheaded test automation frameworks for enterprise options-trading platforms (CRD/ETF-OPT), establishing automated CI/CD quality gates across multi-tier banking systems

## Protean eGov Technologies (formerly NSDL eGov)

*Assistant Manager*

*Mumbai, India*

*Jul 2015 – Sep 2018*

- Managed production deployments and high-throughput transaction pipelines for India's National Pension System (100k+ users) maintaining 99.9% uptime and financial reconciliation.

## PROJECTS

### Job Description Analyzer Agent — Lang Graph multi-agent system

- Architected a two-agent system (JD Parser + Resume Matcher) featuring stateful conditional routing over FAISS vector embeddings. Deployed as a serverless microservice on Google Cloud Run via Terraform and automated GitHub Actions CI/CD.

## Google Cloud Project - video analytics pipeline

- **Serverless Computer Vision Pipeline (GCP & OpenCV):** Engineered an event-driven video processing pipeline using OpenCV, Pub/Sub, Cloud Run, and BigQuery to extract daily frame-level analytics and feature metrics, automating scheduled batch reporting via GCP Cloud Scheduler

## RESEARCH & PATENTS

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- “Hybrid SAMB-GRU: A Regime-Aware Deep Learning Framework for Predictive Maintenance,” IEEE MLISE, published July 2026 — [ieeexplore.ieee.org/document/11607667](https://ieeexplore.ieee.org/document/11607667)
- **Patent Pending (2026):** Neuroidal Network architecture for healthcare ECG analysis, trained via genetic algorithms without backpropagation.

## EDUCATION

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Central Michigan University — M.S., Computer Science

May 2026

Mumbai University — B.E., Information Technology

June 2015

## AWARDS

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- "The Path Finder" Award, JPMorgan Chase, August 2024 (Outstanding Performance & Commitment to Excellence)
- "The Optimizer" Award, JPMorgan Chase, November 2022 & July 2023 (Outstanding Performance & Excellence)
- "Team Player" Award, Oracle (Outstanding Team Contribution)